AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS (THEORY) 2024

Physics Paper II

Time: 1 hour 30 minutes Marks: 35

AKU-EB for Annual Examinations 2024 Teaching & Learning only

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are TEN questions. Answer ALL questions. Questions 9 & 10 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue correcting fluid, or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 3 Marks)

Two-point charges 2.9×10^{-21} C and -2.9×10^{-21} C are separated by a distance of 80 cm.

Calculate the magnitude of electric field intensity at the midpoint of the charges.

(**Note**: Take the value of the constant of proportionality 'k' as 9×10^{9} Nm^2/C^2.)

Q.2. (Total 3 Marks)

A copper wire is hidden beneath a carpet. How can we securely determine the following?

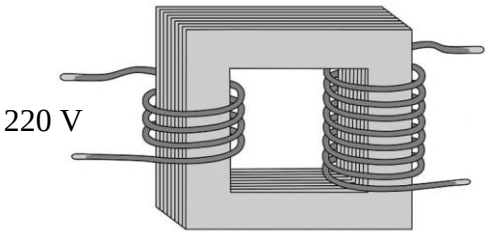
i. Any electrical current flowing through the copper wire (1 Mark)

ii. The direction of magnetic field produced by the electrical current (2 Marks)

AKU-EB Annual Examinations 2024 for Teaching & Learning only

Q.3. (Total 3 Marks)

i. Write the name of the given transformer. (1 Mark)



ii. Calculate the voltage in the secondary coil. (2 Marks)

Q.4. (Total 2 Marks)

How does doubling the frequency of an alternating current affect the reactance of an inductor?

Q.5. (Total 2 Marks)

Define the following terms.

i. Paramagnetism (1 Mark)

ii. Ferromagnetism (1 Mark)

AKU-EB Annual Examinations 2024 for Teaching & Learning only

Q.6. (Total 3 Marks)

i. What is the effect of forward biasing of a diode on the width of depletion region? (1 Mark)

ii. Why is a photodiode operated in reverse biased state? Give TWO reasons. (2 Marks)

Q.7. (Total 2 Marks)

Write any TWO properties of X-rays.

Q.8. (Total 3 Marks)

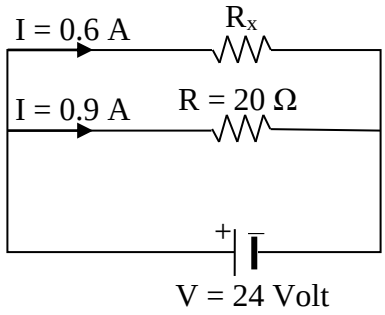
Describe any THREE biological effects of nuclear radiations on the human body.

AKU-EB Annual Examinations 2024 for Teaching & Learning only

Q.9. (Total 7 Marks)

EITHER

a. The given circuit diagram shows two resistors are connected in parallel with a battery.



Calculate the following.

i. The terminal voltage of the circuit.
ii. The unknown resistance (R_x).
iii. The internal resistance of the battery.
iv. The power delivered by the battery.

OR

b. Two colour code resistors exhibit the following colours.

Resistor (a)	Resistor (b)
Band 1: Yellow	Band 1: Violet
Band 2: Violet	Band 2: White
Band 3: Orange	Band 3: Red
Band 4: Gold	Band 4: Silver

Find the value of resistance for each resistor.

Annual Examinations 2024 AKU-EB for Teaching & Learning only

AKU-EB
Annual Examinations 2024
for Teaching & Learning only

Q.10. (Total 7 Marks)

EITHER

a. Describe Albert Einstein's explanation about photoelectric effect and its photoelectric equation.

OR

b. Describe the laws governing black body radiation and explain its FOUR disadvantages.

AKU-EB Annual Examinations 2024 for Teaching & Learning only

END OF PAPER

Please use this page for rough work

AKU-EB
Annual Examinations 2024
for Teaching & Learning only